

Federated O-RAN Slice SLA Prediction: A Cross-Architecture Empirical Benchmark on Colosseum/ColO-RAN

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Abstract—Federated learning (FL) on O-RAN edge gNBs is the natural paradigm for slice-level SLA-violation forecasting because per-user telemetry cannot be centrally pooled. Despite a decade of FL benchmark research, the deployment design space — sequence model architecture, federation algorithm, client-heterogeneity partition, commodity edge hardware — has been studied one axis at a time, leaving the joint behaviour under realistic O-RAN traffic characterised only anecdotally. This paper presents the first cross-architecture empirical FL benchmark on the publicly-available Colosseum/ColO-RAN testbed: 3 sequence backbones (LSTM, Mamba, Spiking-SSM) $\times$ 5 algorithms (FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn) $\times$ 6 partitions (natural-by-BS + Dirichlet $\alpha \in \{0.05, 0.10, 0.50, 1.00, 5.00\}$) $\times$ 10 seeds = 900 cells, with NVML idle-baseline-subtracted training-energy measurement on RTX 4080 commodity hardware, augmented by a 15-cell controlled random-split ablation on 4 $\times$ Tesla V100-SXM2-32GB to probe the leading mechanism hypothesis. We report four empirical findings: (1) client heterogeneity in O-RAN is structurally helpful, not harmful — the natural base-station partition uniformly outperforms every parametric Dirichlet α across all 3 architectures $\times$ 5 algorithms, inverting the standard FL-benchmark assumption; (2) the algorithm-design space is flat at FedAdam-saturated headroom ($\leq +0.016$ AUC over FedAvg), bounding the room available to recent sharpness-aware methods; (3) selective-state-space backbones interact catastrophically with control-variate algorithms at moderate partition skew (Mamba $\times$ SCAFFOLD $\times$ $\alpha \in \{0.10, 0.50\}$); (4) architecture choice traverses 10 $\times$ in commodity-GPU energy and an order-of-magnitude larger AUC range than algorithm choice, dwarfing the algorithmic-leverage scale. We provide a reference benchmark with paired-bootstrap CI_{95} over 270 paired distributions, NVML-instrumented per-cell energy, and pre-registered statistical protocol; the artefact is released under Apache-2.0 with Croissant dataset metadata and a Zenodo DOI on paper acceptance.

Index Terms—Federated learning, O-RAN, slice management, SLA prediction, state-space models, spiking neural networks, NVML energy measurement, paired-bootstrap inference.

I. INTRODUCTION

OPEN Radio Access Networks (O-RAN) decomposes the cellular base station into commodity hardware components controlled by ML-driven xApps and rApps running on disaggregated RAN Intelligent Controllers (RICs) [1]. Slice-level Service Level Agreement (SLA) violation forecasting is one near-RT RIC xApp pattern of practical interest: each gNB simultaneously serves heterogeneous traffic slices (the ColO-RAN release exercises eMBB constant-bitrate, MTC

Poisson, and URLLC Poisson — see Section ??), and proactive forecasting of next-second SLA risk informs near-RT RIC resource-allocation decisions within its 10 ms – 1 s control loop. Because edge gNBs accumulate per-user telemetry that cannot be centrally pooled, federated learning (FL) is the natural training paradigm: each gNB trains a local sequence model on its own traces and shares only weight updates with the FL aggregator. We place the aggregator in the non-RT RIC / SMO orchestration layer (≥ 1 s timescale fits our 100-round federated training cadence per Section ??); a near-RT-RIC-internal xApp aggregator would alternatively be possible if cross-cell xApp messaging supports the federation protocol — we do not engage with that deployment alternative beyond noting it.

Standard FL practice — codified by a decade of toy-image benchmarks — treats client heterogeneity as a *wound* to be healed: lower Dirichlet α (more skew) drives lower accuracy, motivating sharpness-aware methods, control variates, and proximal regularizers. **In O-RAN slice SLA prediction, this premise inverts.** When clients are the *natural* base stations of a Colosseum/ColO-RAN testbed (one client per gNB), the federated average across 7 base-station-natural clients consistently outperforms every uniform-Dirichlet partition of the same training data on the same model + algorithm. Empirically (Section ??, all values are out-of-sample test AUC averaged over 10 seeds), the natural-by-BS partition achieves AUC 0.913–0.918 on LSTM (5-algo range; $\Delta \approx 0.005$), 0.8998–0.9186 on Mamba (with SCAFFOLD as the lower outlier and the other four algorithms in 0.911–0.919), and 0.840–0.856 on Spiking-SSM, **strictly above the AUC obtained at any parametric Dirichlet α** for every (arch, algo) pair, and monotonically increasing as $\alpha \rightarrow 0$ within the Dirichlet sweep. The $\alpha = 5.0 \rightarrow \alpha = 0.05$ AUC gap is ≈ 0.10 – 0.12 on the dense LSTM/Mamba backbones; on Spiking-SSM the gap is smaller (≈ 0.022) but still positive, and across all five algorithms heterogeneity is associated with higher AUC, not lower. The mechanism is open — natural-by-BS is *not* the low- α limit of the Dirichlet sweep (the two partition families redistribute rows along orthogonal axes, see Section ??) — and we present a controlled `random_split` ablation in Section ?? (15 cells $\times$ 3 architectures on 4 $\times$ Tesla V100-SXM2-32GB) showing that any partition that breaks bs grouping (whether by per-slice Dirichlet redistribution at $\alpha = 5.00$ or by uniform random shuffling) collapses AUC to a tight band ~ 0.18 below natural-by-BS, and is statistically equivalent to Dirichlet $\alpha = 5.00$ within seed σ . The mechanism direction (preservation of bs

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grouping) is therefore confirmed; the precise dataset axis the model exploits via bs grouping (channel-state features, see Section ??) remains under further investigation.

Despite the explosion of FL benchmarks since 2018, the deployment design space for FL-based slice SLA prediction remains under-characterized. The space spans four axes:

- 1) **Sequence model architecture** — recurrent (LSTM), structured state-space (Mamba), or bio-inspired sparse spiking variants (Spiking-SSM).
- 2) **Federation algorithm** — FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn (and the more recent sharpness-aware family discussed in Section ??).
- 3) **Client heterogeneity** — including the natural base-station partition (one gNB $\Rightarrow$ one client) and parametric Dirichlet stress $\alpha \in \{0.05, 0.10, 0.50, 1.00, 5.00\}$ over slice IDs.
- 4) **Commodity hardware** — consumer-tier edge GPUs (RTX-class) at the gNB rather than data-center accelerators.

Existing benchmarks address each axis in isolation. NeurIPS-class FL benchmarks evaluate algorithm $\times$ heterogeneity on toy image data [2], [3], [4], [5]. Spiking-NN benchmarks compare architectures in centralized regimes on neuromorphic-friendly datasets [6]. FL $\times$ wireless papers focus on a single architecture [7], [8], [9]. The implicit assumption underlying single-axis recommendations: **best architecture $\times$ best algorithm $\times$ representative $\alpha \times$ commodity hardware = best deployment**.

This paper tests the assumption empirically. We construct the first comprehensive 4-axis FL benchmark on the publicly-available Colosseum/CoLO-RAN testbed dataset [1], [10], sweeping $\{\text{LSTM, Mamba, Spiking-SSM}\} \times \{\text{FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn}\} \times \{\text{natural-by-BS, Dirichlet } \alpha \in \{0.05, 0.10, 0.50, 1.00, 5.00\}\} \times 10 \text{ seeds} = 900 \text{ cells (90 groups, all 10 seeds populated), with NVML-instrumented per-round GPU energy measurement on RTX 4080 (sm_89) commodity hardware. Statistical inference uses paired-bootstrap } CI_{95} \text{ on per-seed AUC deltas; Wilcoxon } p\text{-values are reported uncorrected per finding (with family-wise Bonferroni discussed in Section ?? as a supplementary check).}$

Contributions:

- 1) **Inverted heterogeneity in O-RAN slice federation:** against the standard FL-benchmark assumption, the natural base-station partition uniformly outperforms every parametric Dirichlet α in our sweep across all 3 architectures and 5 algorithms. We do not claim a closed-form mechanism — Section ?? reports a controlled `random_split` ablation that tests the leading “bs-conditioned signal preservation” hypothesis — but the empirical pattern is the first published characterization on Colosseum/CoLO-RAN and is deployment-relevant regardless of which mechanism candidate ultimately survives.
- 2) **Algorithm-design space is flat in this regime; FedAdam achieves the empirical maximum in our 5-algorithm baseline:** across Dirichlet $\alpha \in \{0.05..5.0\}$,

the spread between the best and worst of $\{\text{FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn}\}$ on a fixed (arch, partition) cell is 0.022–0.031 AUC on LSTM/Spiking and 0.078 on Mamba (driven by a single SCAFFOLD outlier). FedAdam consistently leads FedAvg via paired-bootstrap CI_{95} (LSTM Dirichlet-averaged $\Delta = +0.0080$, range $+0.0048.. +0.0128$, all 5 cells significant; Mamba Dirichlet-averaged $\Delta = +0.0062$, range $-0.0021.. +0.0097$, 4 of 5 cells significant; Spiking Dirichlet-averaged $\Delta = +0.0164$, range $+0.0108.. +0.0267$, 4 of 5 cells significant). The mechanism analysis in Sections ?? + ?? + ?? argues that LookAhead-style server-side aggregation (FedSWA / FedSCAM) is *unlikely to systematically exceed* this empirical maximum because of the variance-estimation gap between Adam-style and LookAhead-style server steps (predictive claim, not theorem); we accordingly dismiss FedSWA [11] and FedSCAM [12] via mechanism analysis rather than benchmark them. We do not benchmark FedBN [13] either; Section ?? L2 discusses why the algorithm-flatness ranking is unlikely to be overturned by FedBN given its reported gain envelope on related cellular feature-skew tasks.

- 3) **Architecture $\times$ algorithm catastrophic interaction (Mamba $\times$ SCAFFOLD $\times \alpha \in \{0.10, 0.50\}$):** SCAFFOLD on Mamba at $\alpha = 0.10$ yields AUC 0.7609 ± 0.0830 — std amplified $\sim 3\times$ over (Mamba, FedAvg) at the same α (0.0251) and $\sim 10\times$ over (Mamba, FedAvg) at $\alpha = 5.0$ (0.0073). The deployment warning is concrete: control-variate methods interact pathologically with selective-scan SSMs under high heterogeneity, even as both components individually behave well on neighboring cells.
- 4) **Architecture leverage dominates algorithm leverage (paired-bootstrap CI_{95}) on RTX 4080:** on the energy-AUC Pareto front (Section ??), architecture choice spans $10\times$ in NVML-measured model-attributable energy on RTX 4080 (LSTM $\sim 3.5\text{ kJ} \rightarrow$ Mamba $\sim 10.5\text{ kJ} \rightarrow$ Spiking $\sim 41\text{ kJ}$ per cell). Holding (algorithm, IID partition) fixed, the paired Spiking – LSTM AUC delta is concentrated at -0.061 to -0.074 across all 5 algorithms (all 5 cells significant at $p < 0.005$), substantively dwarfing the FedAdam – FedAvg algorithmic gap of $+0.006$ to $+0.016$ on the same architectures (contribution 2). The $10\times$ span is hardware-specific: the same architectures on a sparsity-aware accelerator (Loihi-2, IBM NorthPole) would yield a substantially smaller LSTM-vs-Spiking energy ratio (Section ?? + Section ?? L1, L6). Operator decisions on commodity edge GPU should center architecture-energy trade-off, not algorithm breadth.
- 5) **Open reproducible reference benchmark:** spec-driven YAML pipeline with pre-flight algorithm validation and `--skip-completed` resumability; paired-bootstrap statistical pipeline; NVML idle-baseline-subtracted training energy; Croissant metadata; Zenodo DOI on acceptance; Apache-2.0 license (preregistered). The full 4-axis table maps deployment configurations $\rightarrow$ (AUC, F1, energy,

paired CI_{95}) over 90 (arch $\times$ algorithm $\times$ partition) groups, enabling lookup-style operator decisions on Colosseum/ColO-RAN.

The remainder of the paper is organized as follows. Section ?? reviews related work across the four axes. Section ?? describes the Colosseum/ColO-RAN dataset and our preprocessing pipeline. Section ?? details the architecture, algorithm, and partition methodology. Section ?? enumerates the reproducibility infrastructure and our Croissant-aligned data release. Section ?? reports results across the 90 (arch $\times$ algorithm $\times$ partition) groups with paired-bootstrap CI_{95} . Section ?? discusses the mechanism finding and articulates the applicability boundary on commodity edge hardware. Section ?? enumerates limitations. Section ?? concludes.

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